



GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE: CM5469

COURSE TITLE: DATABASE MANAGEMENT SYSTEM

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V		3	-	4	7	16	48	-	64

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External Assessment: Practical based \$: Online Examination

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1. RATIONALE:

Each and every organization like shopping mall, hospital, banking, institutes, industry need to share huge amount of data in effective manner. This course aims to develop skills in students to create, store, modify, manage and extract information from a database. Database system can be used as a backend for developing database application.

2. COURSE OUTCOMES (COs)

The theory, practical experiences and soft skills associated with this course are to be taught and implemented, so those students demonstrate the following:-

- 1) Design Normalized database on given database.
- 2) Create and Manage Database using SQL command.
- 3) Write PL/SQL code for given database.
- 4) Apply triggers on database also create procedure and function according to conditions.
- 5) Apply security and confidentiality on given database.
- 6) Identify the basic issues of transaction processing, concurrency control, recover techniques.

3.DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr.	Marks
Unit I Database System Concept	1a. State the importance of DBMS over the file processing in the given situation. 1b. Describe the overall structure of the given DBMS. 1c. Identify the relevant database model in the given situation. 1d. Draw the ER diagram of the given database and identify the relationship between the entities.	1.1 Concept of Data, database, DBMS, advantages of DBMS over file processing system, Application of database 1.2 Three level architecture for Database system. 1.3 Data abstraction: Different levels of data abstraction, Instances and schema, data independence-Logical and Physical Independence. 1.4 Overall structure of DBMS. 1.5 Data Modeling: Record base logical model –Relational, Network, Hierarchical. 1.6 Data Modeling using ER model: Entity Relationship model, Strong entity set, Weak entity set, Types of attributes E-R diagram.	1	08	08
Unit II Relational data model	2a. Explain the concept of RDBMS also appropriateness for the given problem. 2b. Design Normalized database structure in given problem. 2c. Design SQL queries to create Relational Database and apply in the given data constraints. 2d. Identify the operators for queries implementation of the given problem.	2.1 Fundamentals of RDBMS- records, fields, data types, tables and database. 2.2 Concept of RDBMS, E.F.codd's rule for RDBMS, key concepts-candidates key, Primary key and foreignkey. 2.3 Normalization: Normalization concepts, Need of Normalization, Types of Normalization-1NF, 2NF, 3NF. 2.4 Introduction to Structure Query Language, Data Type in SQL, Components of SQL- DDL, DML, DCL, DQL 2.5 DDL Commands: CREATE, ALTER, DROP, TRUNCATE, DESCRIBE, RENAME 2.6 Data Integrity Constraint: Types of Data Integrity Constraints: I/O constraint- Primary key, foreign key,	2	12	16

		Unique key Constraints, Business Rule Constraints- Null, Not Null and Check Constraints. 2.7 DML Commands: INSERT, UPDATE, DELETE 2.8 DC Commands: COMMIT, SAVEPOINT, ROLLBACK, GRANT and REVOKE. 2.9 DQL Command: SELECT 2.10 SQL Operators: Arithmetic operators, Comparison operators, Logical operators, set operators, Range Searching Operators-Between, Patterns matching Operators-Like.			
Unit III Interactive SQL and Advance SQL:SQL Performance Tuning	3a. Write the given queries using the relevant functions. 3b. Write a query to combine the given multiple table using join. 3c. Design SQL Queries to implement VIEWS on given tables. 3d. Apply and drop INDEXES and SYNONYM on the given table.	3.1 In-built Functions: String, Arithmetic 3.2 Date and Time, Aggregate Functions. 3.3 Queries using Group by, having, and Order by clause, joins-Inner and Outer joins, sub queries. 3.4 Views: Concept of views, The Create views command, defining views command, Views and Joins, Views and Sub queries, Dropping Views. 3.5 Subsequence: Creating Sequence, Altering Sequences, Dropping Sequences. 3.6 Indexes: Indexes type, Creating of an Indexes: Simple Unique and Composite Indexes, Dropping Indexes 3.7 Synonyms: Creating Synonyms, Dropping Synonyms	3	06	12

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr.	Marks
Unit IV PL/SQL PROGRAMMING	4a. Write simple PL/SQL Code using control Structure and handle various exceptions in the given situation. 4b. Create cursor for retrieving multiple records in the given situation. 4c. Create and Execute stored procedure and functions in the given situations. 4d. Create and apply Database trigger using PL/SQL In the given situation.	4.1 Introduction to PL/SQL, Advantages of PL/SQL, The PL/SQL Block Structure, PL/SQL execution environment, PL/SQL data Type, variable, constant. 4.2 Control Structure: Conditional Control, Iterative Control, Sequential Control. 4.3 Exception handling: Predefined Exception, User Defined Exception. 4.4 Cursors: Implicit and Explicit Cursors, Declaring, Opening and Closing a Cursors, Fetching a record from Cursors, Cursors for loops, Parameterized Cursors. 4.5 Procedure: Advantages, Creating, Executing and Deleting a stored procedure. 4.6 Functions: Advantages, Creating, Executing and Deleting Functions. 4.7 Database Triggers: Use of Database Triggers, How to apply Database Triggers, Types of Triggers, Syntax for Creating Triggers, Deleting Trigger.	4	10	12
Unit V Database Security	5a. Provide the security to the given database by assigning various privileges to user. 5b. Create and Manage the given database user. 5c. Explain the Advantages of Database Backup and recovery in given situation.	5.1 Database security: Introduction to database security, Data security Requirements, Types of Database Users-Creating, altering and Deleting Users. 5.2 Protecting the data within database-Database privileges: System privileges, Object privileges, Granting and revoking privileges: Grant and Revoke command. 5.3 Database Backup-Types of failures, Causes of failures, Database Backup introduction, Type of Database Backups-Physical and Logical. 5.4 Database Recovery-Recovery concept, Recovery Techniques-	5	06	10

		Roll Forward, Roll Back.			
Unit VI Transaction processing	6a. Explain the importance of Transaction 6b. Describe Locking protocols 3) State Deadlock and Deadlock handling Techniques.	6.1 Concept of transaction, States of transactions, Concurrent Executions, Serializability , Transaction Definition in SQL. 6.2 Lock based protocols : share and exclusive models, Protocols: • 2 phase locking • Time-Stamp based • Validation based 6.3 Deadlock handling, Deadlock prevention, Detection & recovery.	6	06	10

4.LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (Pros)	CO NO.
1	Design and draw Entity Relation Diagrams on given data.	1
2	Perform following in GUI based database software using GUI only i) Create Database. ii) Create tables and assign primary key. iii) Modify the table structure-add column, change the data type of column, delete the column from table. iv) Insert, delete, and update the record from the table. v) Retrieve data from the table according to conditions given.	2
3	Perform following in GUI based database using GUI only i) Apply given validation on table and set error messages. ii) Set default value for column. iii) Set and remove database password.	2
4	Design and Normalize database up to three normalization form	2
5	Create and Execute DDL commands using SQL. i)CREATE ii) ALTER iii) TRUNCATE iv) DROP v)RENAME	2
6	Apply following Integrity constraints to table: i) Primary key, Foreign key, Unique key constraints, Null, Not null and Check constraints.	2
7	Create and Execute DML commands using SQL.	2
8	Write Queries using following operators: Arithmetic Operators, Comparison Operators, Logical Operators,	2
9	Write Queries using following operators: Word Comparison Operator, Range Searching operators-Between, Patterns matching operators-Like.	2
10	Write Queries using Set Operators	
11	Write a queries using following Functions: String, Arithmetic, Date and time	3
12	Write a queries using Aggregate functions.	3
13	Execute Queries using the Select command with Where, Having, Group by and Order by clause.	3

Sr No.	PRACTICAL OUTCOMES (Pros)	CO NO.
14	Execute the queries for implementation of Inner and Outer Join.	3
15	Implement Views: i) Create different views. ii) Insert, modify and delete records through the views. iii) Delete the views.	3
16	Create and Executes Indexes, Sequences and synonyms in SQL.	3
17	Write PL/SQL programs using if then else, for, while, nested loops.	4
18	Write a PL/SQL code to implement implicit and explicit cursors.	4
19	Write PL/SQL programs based on Expectations handling.(Predefined and user-defined exceptions)	4
20	Write a PL/SQL code to create Procedure .	4
21	Write a PL/SQL code to create Functions	
22	Write a PL/SQL code to create triggers on given database.	4
23	Executing DCL commands using SQL i) Create users. ii) Grant privileges to users. iii) Revoke privileges from users.	5
24	Write a PL/SQL program to lock the table in shared mode and exclusive mode	4,6

Note

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain'.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

S. No.	Performance Indicators	Weightage in %
1	SQL queries and PL/SQL programming	60
2	Correctness in Database Design	10
3	Quality result displayed by SQL queries	10
4	Answer to sample Questions	10
5	Submission of report in time	10
Total		100%

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Follow safety practices.
- Practice good housekeeping.
- Demonstrate working as a leader/a team member.
- Follow ethical practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organizing Level' in 2nd year and
- 'Characterizing Level' in 3rd year.

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a. Prepare journal of practical.
- b. Undertake micro-projects.
- c. Prepare a chart to classify Data Structures.
- d. Prepare charts for logical representation of Data Structures.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

These are sample strategies, which the teacher can use to accelerate the attainment of the various learning outcomes in this course:

- a. Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b. Guide student(s) in undertaking micro-projects.
- c. Demonstrate students thoroughly before they start doing the practice.
- d. Encourage students to refer different websites to have deeper understanding of the subject.
- e. Observe continuously and monitor the performance of students in Lab.
- f. 'About **10-15% of the topics/sub-topics** which is relatively simpler or descriptive in nature is to be given to the students for **self-directed learning** and assess the development of the COs through classroom presentations (see implementation guideline for details).

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. In all semesters the micro-project are group-based (5-6 students) to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **16 (sixteen) student engagement hours** during the course. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

- a) Survey on various database system software and compare it.
- b) Design E-R Diagram for Hospital/college/medical/ jewellery shop/Blood bank/Library.
- c) Design Normalized database for Hospital/college/medical/jewelleryshop/Blood bank/Library.
- d) Apply a trigger on given database.
- e) Create procedure and function according to given conditions.
- f) Design scenario and an E-R diagram for IT training group database Project
- g) Any other micro project suggested by subject faculty on similar line.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

The major equipment with broad specification mentioned here will usher in uniformity in conduct of experiments, as well as aid to procure equipment by authorities concerned.

Sr. No.	Equipment Name with Broad Specifications	PrO S. No.
	Computer system (Any computer system which is available in laboratory)	All
	Any GUI based SQL software (MsAccess/MySQL), Oracle, MariaDB, PostgreSQL	
	Any RDBMS software(MySQL/ SQLServer)	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Database System Concept	08	04	04	--
2	Rational Data Model	16	04	08	04
3	Interactive SQL and Advance SQL:SQL performance Tuning	12	--	04	08
4	PL/SQL Programming	12	04	04	04
5	Database security	12	04	04	04
6	Transaction Processing	10	02	04	04
Total		70	18	28	24

R=Remember, U=Understand, A=Apply and above (Bloom's Revised taxonomy)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

S. No.	Title of Book	Author	Publication
1.	Database System Concepts-6e	Silberschatz Abraham, Korth Henry, Sudarshan S.	McGraw Hill Education, New Delhi 2013, ISBN: 978-93-329-0138-4
2.	Database Management System- A practical approach	Chopra, Rajiv	S. Chand and Company, New Delhi 2016, ISBN:978-93-856-7634-5
3.	Fundamentals of Database Systems-6e	Ramez Elmasri Shamkant Navathe	Addison Wesley, New Delhi 2003 ISBN 13: 978-0-136-08620-9
4.	Database Management Systems	Gupta, G. K.	McGraw Hill Education, New Delhi 2013, ISBN: 978-07-107273-1
5.	An Introduction to Database Systems	Date, C. J.	Addison Wesley, New Delhi 2003 ISBN 13: 9780321189561

11. SOFTWARE/LEARNING WEBSITES.

- a. www.tutorialpoint.com
- b. www.wielyindia.com
- c. <http://phindia.com/gupta/chapter/chapter1.jxlf>
- d. www.williamstannings.com
- e. www.nptl.ac.in
- f. <http://www.khanacademy.org/>

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1.	S. S. CHAVHAN	Lecturer in Computer Engineering.	Govt. Polytechnic Amravati
2.	V.R.RATHOD	Lecturer in Computer Engineering	Govt. Polytechnic Amravati
3.	P.P.KATGAONKAR	Lecturer in Information Technology	Govt. Polytechnic Amravati

Government Polytechnic Amravati, Programmed Board of Studies Computer Engineering has approved the above course curriculum on 14/05/2020 is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 14/07/2020

The Governing Body has approved the above course curriculum on.13/08/2021



GOVERNMENT POLYTECHNIC, AMRAVATI
(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)
CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING/
INFORMATION TECHNOLOGY**

COURSE CODE: FC5490

**COURSE TITLE: EMERGING TRENDS IN COMPUTER ENGINEERING/
INFORMATION TECHNOLOGY(For Academic year 2020-2021)**

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V		02	--	01	03	16	32	--	16

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
02	MAX.	35\$	15*	50	25#	25^	100
	MIN.	14		20	10	10	-

@: Internal Assessment #: External Assessment Skill based \$: online examination

(*) Under the Theory PA, Out Of 15 Marks, 10 Marks is the Average of Two Tests and 05 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) -40%.

1. RATIONALE:

Advancements and Applications of Computer engineering and Information Technology are ever changing Emerging trends aims at creating awareness about major trends that will define technological disruption in the upcoming years in the field of Computer engineering and Information Technology. There are some emerging areas expected to generate revenue, increasing demand as IT professionals and open avenues of entrepreneurship.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Describe Artificial Intelligence ,machine learning and deep learning
2. Interpret Internet of Things concepts.
3. Describe Ethical Hacking process and Detect Network Vulnerabilities..